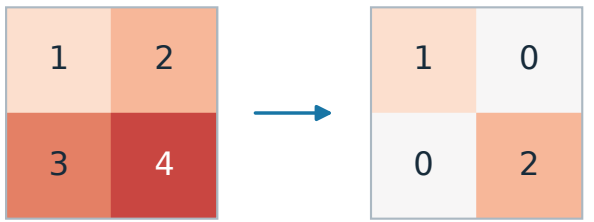
From patch projection to transformer components

Linear patch mixing, input-dependent attention, and complete residual accounting.

1 · Patch embedding



One token channel: weights [1, 0, 0, 0.5]

$$h_t = W_E x_{P_t} + b + p_t$$

$$\Delta h_t = W_E \Delta x_{P_t}$$

Local learning image: [1, 0; 0, 2], sum = 3.

Bias and position stay in the model.

2 · Attention

$$O = A(x)V(x)$$

$$\mathrm{d}O = (\mathrm{d}A)V + A\,\mathrm{d}V$$

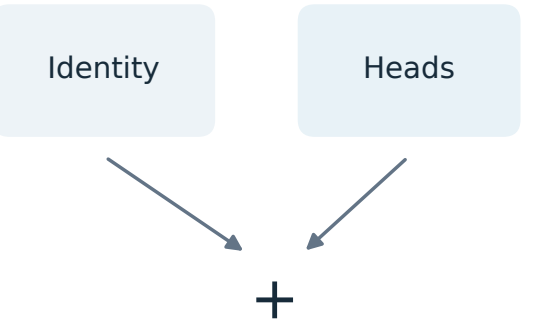
Full model: retain both terms.

Fixed-attention view: retain A dV.

Softmax / LayerNorm / GELU:

use derivatives along the path.

3 · Residual branch



$$\mathcal{L}_{\mathrm{id}} + \sum_h \mathcal{L}_h = \mathcal{L}_{\mathrm{total}}$$

Count each route exactly once.

Seed images are a mathematical representation; autograd evaluates the full input path, not each seed in isolation.